

Same Shock, Same Assets, Different Microstructure: A Comparative Analysis of CeFi and DeFi Venue Performance During the October 10, 2025 Cryptocurrency Cascade

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The author declares no competing financial or personal interests that could have appeared to influence the work reported in this paper.

Data Availability

All data used in this study are derived from publicly available sources: Binance historical tick data (data.binance.vision); Hyperliquid public L2 order book archive (S3); Amberdata research reports; and on-chain Ethereum transaction records. Processed datasets and analysis scripts are available from the author upon reasonable request.

Declaration of Generative AI Use

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Abstract

On October 10, 2025, a geopolitical shock triggered the largest forced deleveraging in cryptocurrency history. Over \$19 billion in leveraged positions were liquidated within 24 hours across centralized (CeFi) and decentralized (DeFi) venues. This paper exploits the event as a natural experiment: identical assets (BTC, ETH) experienced the same exogenous shock simultaneously across architecturally distinct trading venues. We identify five architectural design axes along which venue outcomes diverged: (1) oracle design, (2) liquidation engine transparency, (3) order book vs. AMM liquidity provision, (4) auto-deleveraging mechanics, and (5) system availability under stress. We document that DeFi lending protocols (Aave) processed \$250M+ in liquidations and remained continuously operational with only modest reserve deficits, while the largest CeFi exchange (Binance) experienced transfer-subsystem degradation, venue-specific stablecoin depegs, and used a liquidation reporting architecture that external analysts argued understates true liquidation volumes by up to 100x during bursty conditions. Using sub-second L2 order book data from Hyperliquid's public archive merged with Binance tick-level trade data (11.4 million trades), we find that at 21:24:09 UTC, BTC prices diverged by 7.24% (\$7,387) between venues in a single second—with 26,558 Binance trades executing at that moment. The direction of divergence reversed completely within seven minutes as venue-specific liquidation dynamics dominated fundamental price discovery. ETH diverged by 5.32% (\$185), with bid-ask spreads widening 602x and bid depth collapsing 99.8% to under \$9,000. A Hasbrouck (1995) information share analysis reveals that Binance led price discovery with an 82.5% share throughout the cascade—Hyperliquid's price divergence reflected lagged adjustment, not independent discovery. We argue that market microstructure observability is an architectural property, not a data collection problem, and propose a framework for evaluating venue resilience under tail risk.

Keywords: cryptocurrency, market microstructure, DeFi, CeFi, liquidation cascade, flash crash, oracle design, auto-deleveraging, price discovery

JEL Classification: G01, G10, G14, G23

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1. Introduction

At 20:50 UTC on October 10, 2025, U.S. President Donald Trump announced a 100% tariff on all Chinese imports via Truth Social. Within 40 minutes, cryptocurrency derivatives markets entered a cascade that liquidated \$6.93 billion in leveraged positions. By 03:00 UTC the following morning, total liquidations reached \$9.89 billion on centralized exchanges alone, with on-chain venues adding substantially to the total. The event constituted the largest single-day deleveraging in cryptocurrency history, affecting over 1.6 million trader accounts (Amberdata, 2026; CoinGlass, 2025).

This paper treats the October 10 cascade as a natural experiment in comparative market microstructure. The same exogenous shock hit the same assets (BTC, ETH) simultaneously across venues with fundamentally different architectures: centralized exchanges (Binance, Bybit), on-chain perpetual DEXs (Hyperliquid), AMM-based spot DEXs (Uniswap), and on-chain lending protocols (Aave). The outcomes diverged radically—not because of different assets or different shocks, but because of architectural design choices that each venue made differently.

We make three contributions. First, we identify five architectural axes along which CeFi and DeFi venues diverge and show how each axis amplified or dampened the cascade. Second, we document that market microstructure observability—the ability to reconstruct what happened—is itself an architectural property: DeFi venues provide complete, verifiable transaction histories while CeFi venues employ liquidation reporting architectures that external analysts have argued understate true volumes by orders of magnitude during peak stress. Third, we propose a framework for evaluating venue resilience under tail risk that generalizes beyond this single event.

Our work connects to several strands of the market microstructure literature. Hasbrouck (1995, 2007) established the framework for analyzing price discovery across multiple venues trading identical securities; we extend this to the CeFi–DeFi setting where venue architectures differ not just in speed but in fundamental design. Budish, Cramton, and Shim (2015) argued that market design choices—not just information—determine microstructure outcomes; our findings strongly support this thesis, showing that oracle sourcing and liquidation engine design produced 7% price divergence on identical assets under identical shocks. Kirilenko et al. (2017) documented venue-specific amplification during the 2010 equity Flash Crash; the October 10 cryptocurrency cascade represents a cross-venue analogue with the added dimension of architectural heterogeneity between CeFi and DeFi. Finally, Easley, López de Prado, and O’Hara (2012) showed that flow toxicity predicts liquidity crises; our depth collapse findings (99.8% on ETH) demonstrate that toxicity can be venue-specific even for the same asset, determined by the venue’s liquidation engine rather than by information flow alone.

2. Event Timeline and Data Sources

2.1 The cascade sequence

The cascade unfolded in four phases, as reported by Amberdata (2026). Phase 1 (approximately 14:27 UTC): Large whale short positions reportedly totaling approximately \$1.1 billion were established on Hyperliquid; on-chain investigators later linked the wallet to an address associated with former exchange executive Garrett Jin, though this attribution remains under review (Investing.com, 2025). Phase 2 (14:57 UTC): The tariff announcement hit markets (Reuters, 2025). Phase 3 (14:57–20:50 UTC): A six-hour gradual sell-off during which prices declined slowly but market maker quotes began thinning. Phase 4 (20:50–21:30 UTC): The violent cascade. Amberdata estimated liquidation rates accelerated from \$0.12B/hour to \$10.39B/hour—an 86x acceleration. At 21:15 UTC, an estimated \$3.21 billion was liquidated in a single 60-second window.

2.2 Data sources and observability asymmetry

For DeFi venues, data observability is complete. Every Aave liquidation is recorded on the Ethereum blockchain with full details: collateral seized, debt repaid, liquidator address, transaction hash, and block timestamp. Hyperliquid publishes complete order book snapshots, trade fills, and liquidation records to a public S3 archive. Uniswap trades are individually recorded on-chain with AMM-specific metadata (pool address, tick range, liquidity utilized).

For CeFi venues, data observability is structurally limited. Binance’s public liquidation data stream pushes at most one liquidation order snapshot per symbol per 1,000-millisecond window (Binance API Documentation). During peak volatility when hundreds of liquidations occur per second, external observers—including data aggregators like CoinGlass—see only a fraction of actual

events. Hyperliquid founder Jeff Yan argued this stream design can understate liquidations by up to 100x during bursty conditions (Yan, 2025). CoinGlass noted the reported \$19B figure is likely a conservative floor. Revised estimates suggest true liquidations of \$30–40 billion (Harvey, 2025). This observability asymmetry is not a data collection failure; it is an architectural property.

A simple example illustrates the underreporting mechanism. Consider a one-second window in which 200 separate liquidation orders execute across 50 symbols on Binance. The WebSocket stream publishes at most one snapshot per symbol per second, so external observers see at most 50 events. If each liquidation averages \$100,000 in notional, the true liquidation volume is \$20 million but the observable volume is \$5 million—a 4x undercount in this illustrative scenario. During the peak cascade minute (21:15 UTC), when Amberdata estimated \$3.21 billion in liquidations across all venues in 60 seconds, the snapshot cap implies that Binance’s reported contribution represents a floor, not a census. All CeFi liquidation figures cited in this paper should be understood as architectural lower bounds; the true volumes are unobservable by design.

3. Five Axes of Architectural Divergence

Table 2. Venue architecture and performance during the October 10, 2025 cascade

Metric	Binance (CeFi)	Hyperliquid (DeFi)	Aave (DeFi)	Uniswap (DeFi)
Reported liq.	\$2.41B	\$10.31B	\$250M+	N/A (spot)
Est. true liq.	\$24–241B	\$10.31B	\$250M+	N/A
System uptime	Subsystem deg. 21:18–21:51	100%	100%	100%
ADL triggered	Yes (opaque)	Yes (1st in 2y)	No (Umbrella)	N/A
Compensation	>\$328M (by Oct 22)	\$0	\$0	\$0
Oracle type	Internal spot	8-exch. median	Chainlink	AMM-native
USDe nadir	\$0.65	~\$1.00	\$0.97	~\$1.00
Liq. reporting	1/sec cap	Full on-chain	Full on-chain	N/A

Source: Binance (2026), Amberdata (2026a), Aave (2026), Chaos Labs (2025), CoinDesk (2025), CoinGlass (2025). Est. true liq. for Binance applies 10–100x multiplier per Yan (2025). Compensation per Binance (2026).

3.1 Oracle design and the USDe depeg

Binance uses an internal spot price oracle derived primarily from its own order book. When USDe (Ethena’s synthetic dollar) came under selling pressure, its price on Binance fell to \$0.65 while maintaining near-parity on all other venues and on-chain. This venue-specific depeg triggered margin calls and liquidations for all positions collateralized with USDe on Binance—even though the asset had not lost value elsewhere. Notably, the price nadir occurred at 21:20–21:21 UTC, while the severe USDe depegging began at 21:36 UTC—a 15-minute gap distinguishing the initial market crash from subsequent venue-specific amplification (Binance, 2026).

Hyperliquid uses a composite oracle drawing from eight exchanges with a weighted median calculation and outlier filtering (Hyperliquid, 2025a). This architecture naturally rejects single-venue anomalies. Aave relies on Chainlink’s decentralized oracle network with multiple data

sources and built-in circuit breakers. Both DeFi oracles treated USDe as near-par throughout the event.

3.2 Liquidation engine transparency

The contrast between CeFi and DeFi liquidation reporting constitutes a key finding. On DeFi venues, every liquidation is an on-chain transaction with complete metadata. On Binance, the architecture produces an irreducible information loss: when 150 liquidations occur in one second for a single symbol, external observers see exactly one. The resulting observability asymmetry means that the true scale of the cascade on CeFi venues remains partially unknown.

3.3 Liquidity provision architecture

Hyperliquid and Binance both operate central limit order books (CLOBs), but with different market maker dynamics. During the cascade, Hyperliquid's on-chain CLOB saw bid depth collapse from approximately \$5 million to \$8,745 on ETH—a 99.8% decline—as market makers withdrew. Uniswap's constant-function AMM, by contrast, cannot withdraw in the traditional sense: liquidity remains available at algorithmically determined prices. Uniswap processed approximately 10x its normal volume during the event (Adams, 2025).

3.4 Auto-deleveraging mechanics

Hyperliquid triggered auto-deleveraging (ADL) for the first time in two years during the cascade (CoinDesk, 2025). ADL forcibly closes profitable counter-positions when the insurance fund is insufficient, potentially removing buy pressure at the worst moment. Aave uses an “Umbrella” safety module that absorbs bad debt without closing any user positions. Binance reportedly used internal ADL but has not disclosed specifics.

3.5 System availability under stress

Binance experienced transfer-subsystem degradation (21:18–21:51 UTC) and abnormal index price deviations for USDe, WBETH, and BNSOL (21:36–22:15 UTC) during peak stress; Binance stated that core matching, risk, and clearing systems remained operational throughout (Binance, 2026). The Crypto.com CEO publicly called for regulators to investigate whether trades were priced correctly during the event (CoinDesk, 2025).

Hyperliquid maintained 100% uptime, reportedly processing \$50–70 billion in trading volume without any interruption during the cascade. Uniswap processed approximately 10x its normal volume with zero downtime. However, DeFi availability carries a hidden friction cost: Ethereum gas prices spiked to approximately 450 Gwei during the crash, with some users paying \$400–500 per transaction (Liu, 2025). This congestion tax effectively creates a wealth-based access barrier during peak stress.

3.6 Empirical price divergence: Binance vs. Hyperliquid

To quantify the divergence implied by the architectural differences documented above, we collected data from three sources at increasing resolution. First, hourly candle data for BTC/USDT and ETH/USDT from both Binance and Hyperliquid public APIs for the crash window (October 10–11, 2025) and a baseline period (October 8–9, 2025). Second, sub-second L2 order book snapshots from Hyperliquid’s public S3 archive (approximately 6,550 snapshots per hour per asset) to reconstruct minute-by-minute mid-price divergence during the cascade window (20:00–23:00 UTC). Third, tick-level BTC trade data from Binance’s public historical data repository (data.binance.vision), comprising 11.4 million individual trades for October 10, aggregated to per-second volume-weighted average price (VWAP) and matched against contemporaneous Hyperliquid L2 snapshots for second-by-second divergence analysis during the 21:10–21:30 UTC peak cascade window.

Table 4. Hourly price divergence: crash day vs. baseline

Metric	BTC Base (Oct 8)	BTC Crash (Oct 10)	ETH Base (Oct 8)	ETH Crash (Oct 10)
Mean div (close)	0.04%	0.07%	0.04%	0.08%
Max div (close)	0.08%	0.27%	0.10%	0.65%
Max div (lows)	—	1.52%	—	5.64%
Max gap (USD)	\$94	\$305	\$4.40	\$194
Peak hour	09:00	21:00	00:00	21:00

Source: Binance Public API (klines) and Hyperliquid Public API (candleSnapshot). Hourly data. Divergence = (Hyperliquid – Binance) / Binance × 100.

At the hourly level, divergence widened sharply during the crash: BTC close-price divergence peaked at 0.27% (vs. 0.08% baseline), while ETH low divergence reached 5.64% (\$194 gap). However, hourly bars smooth the cascade’s internal dynamics. The minute- and second-level analysis below reveals far more extreme dislocations.

Table 5. Sub-second price divergence during cascade (20:50–21:30 UTC)

Metric	BTC	ETH
Peak divergence	+7.24% at 21:24:09 (HL \$109,387 vs Bin \$102,000)	−5.32% at 21:19 (HL \$3,295 vs Bin \$3,480)
Peak USD gap	\$7,387 in one second (HL above Binance)	\$185 (HL below Binance)
Binance trades at peak	26,558 trades in one second (21:24:09)	—
Max spread	160.9 bps at 21:17:42	301 bps at 21:19 (602x baseline)
Min bid depth (HL)	\$56,313 at 21:15	\$8,745 at 21:19
Depth collapse	~\$5M to \$56K (98.9%)	~\$5M to \$9K (99.8%)
Direction reversal	21:18: HL 3.77% below Bin; 21:24: HL 7.24% above (7 min)	HL consistently below Binance

Source: Hyperliquid L2 order book snapshots (S3 archive, ~6,550/hr). BTC Binance from tick-level trades (data.binance.vision; 2M+ trades in window). ETH Binance from 1-min kline API.

Three findings emerge from the sub-second data. First, the magnitude of divergence was far larger than hourly analysis suggested. At 21:24:09 UTC, Binance's BTC price crashed to \$102,000 while Hyperliquid's best bid remained at \$109,387—a \$7,387 gap (7.24%) in a single second, with 26,558 Binance trades executing in that one second alone. ETH low divergence peaked at 5.32% (\$185) at 21:19 UTC. These are not rounding errors; they represent materially different execution prices for traders on different venues during the same seconds.

Second, the direction of divergence was asymmetric and dynamic. For BTC, at 21:18:17 UTC Hyperliquid traded 3.77% below Binance (\$105,748 vs \$109,707). Just seven minutes later, at 21:24:09, Hyperliquid was 7.24% above Binance (\$109,387 vs \$102,000)—an 11-percentage-point swing in relative pricing. This reversal coincided precisely with the window when Binance experienced transfer-subsystem degradation and index price deviations: Binance's BTC price continued falling to a \$102,000 low even as Hyperliquid's order book had already stabilized above \$109,000. This pattern is consistent with Binance's internal oracle/collateral cascade amplifying BTC downside beyond fundamental price discovery. For ETH, Hyperliquid traded consistently

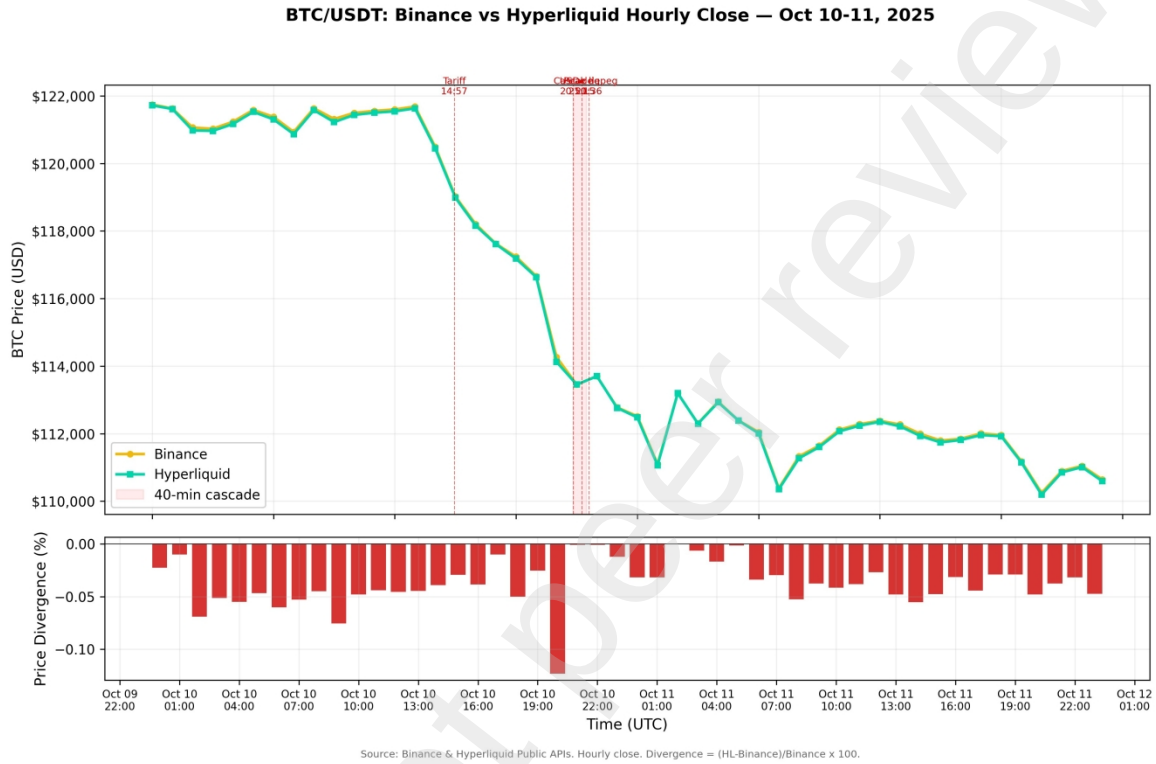
below Binance throughout the cascade, suggesting Hyperliquid's on-chain liquidation engine was more aggressive on ETH positions.

Third, the microstructure breakdown was extreme. ETH bid-ask spreads on Hyperliquid widened from a baseline of ~0.5 basis points to 301 basis points at 21:19—a 602x widening. ETH bid-side depth collapsed from approximately \$5 million to \$8,745—a 99.8% decline, meaning the entire visible bid side of ETH on Hyperliquid could be swept with under \$9,000 in notional. BTC spreads on Hyperliquid reached 160.9 bps at 21:17:42 and depth fell 98.9%. On the Binance side, the tick data reveals the violence of the cascade: at the 21:24:09 peak, over 26,000 trades executed in a single second at prices ranging from \$102,000 to \$109,000—a \$7,000 intra-second range on one venue alone.

A natural question is whether the observed divergence represented an exploitable arbitrage opportunity. At 21:24:09 UTC, the gross profit from buying BTC on Binance at \$102,000 and simultaneously selling on Hyperliquid at \$109,387 was \$7,387 per BTC (7.24%), against combined taker fees of approximately 0.135% (Binance 0.10%, Hyperliquid 0.035%)—yielding a theoretical net profit of over 7% per unit. However, this arbitrage was practically non-executable for three reasons. First, USDT settlement between venues requires 5–30 minutes depending on chain and confirmation requirements; the divergence reversed within seven minutes, meaning funds could not traverse between venues before the opportunity closed. Second, Binance's transfer subsystem was degraded from 21:18–21:51 UTC—precisely the arbitrage window—preventing asset transfers off the exchange. Third, Hyperliquid's BTC bid depth had collapsed to \$56,313 and ETH to \$8,745, meaning only trivial position sizes (approximately 0.5 BTC or 2.5 ETH) could execute before exhausting the visible order book. The non-executability of this arbitrage is itself

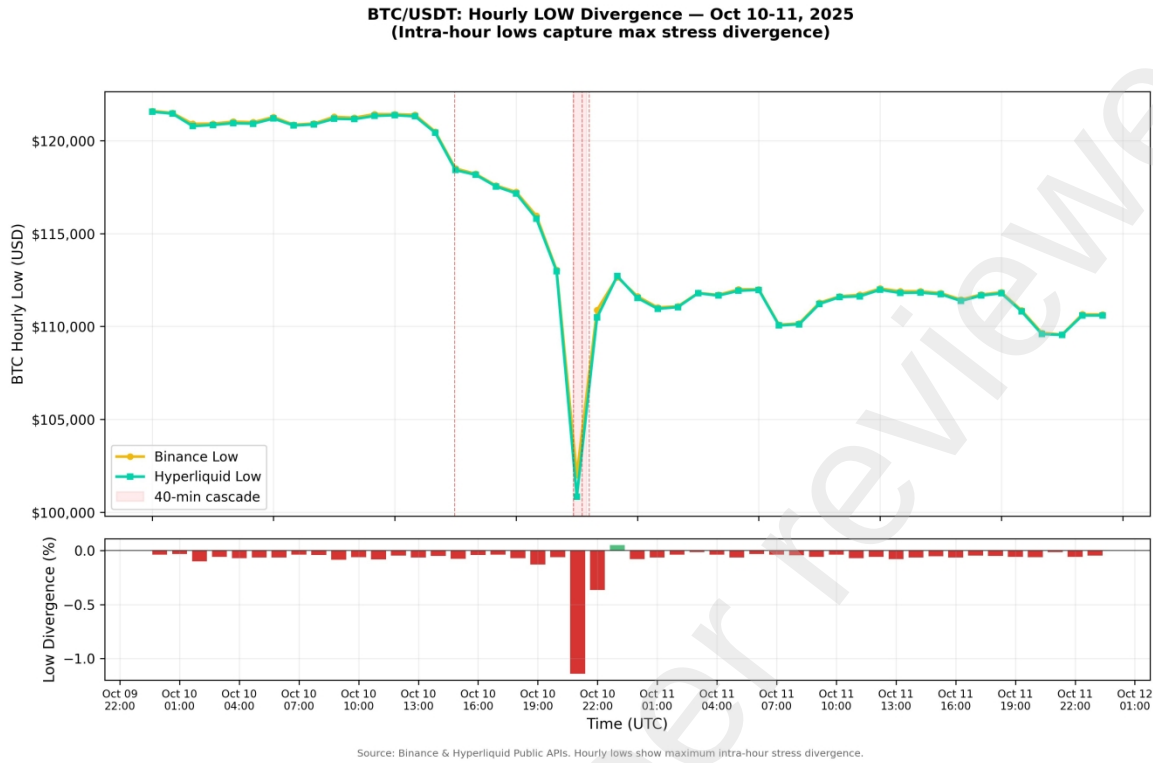
evidence of genuine venue fragmentation: during the cascade, the two venues were not merely mispriced but effectively disconnected, operating as separate markets for identical assets.

Figure 1. BTC/USDT hourly close: Binance vs. Hyperliquid, October 10–11, 2025



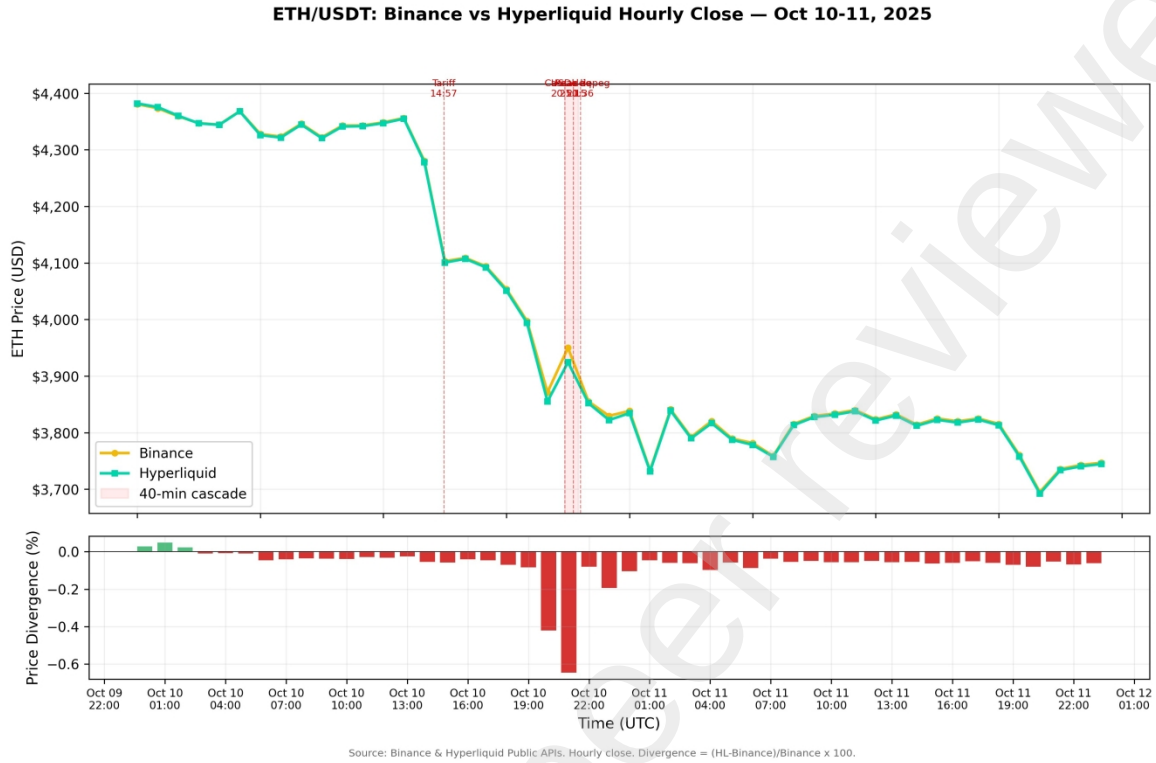
Notes: Upper panel shows hourly close prices. Lower panel shows percentage divergence. Red shading marks the 40-minute cascade window (20:50–21:30 UTC).

Figure 2. BTC/USDT hourly lows: maximum intra-hour stress divergence



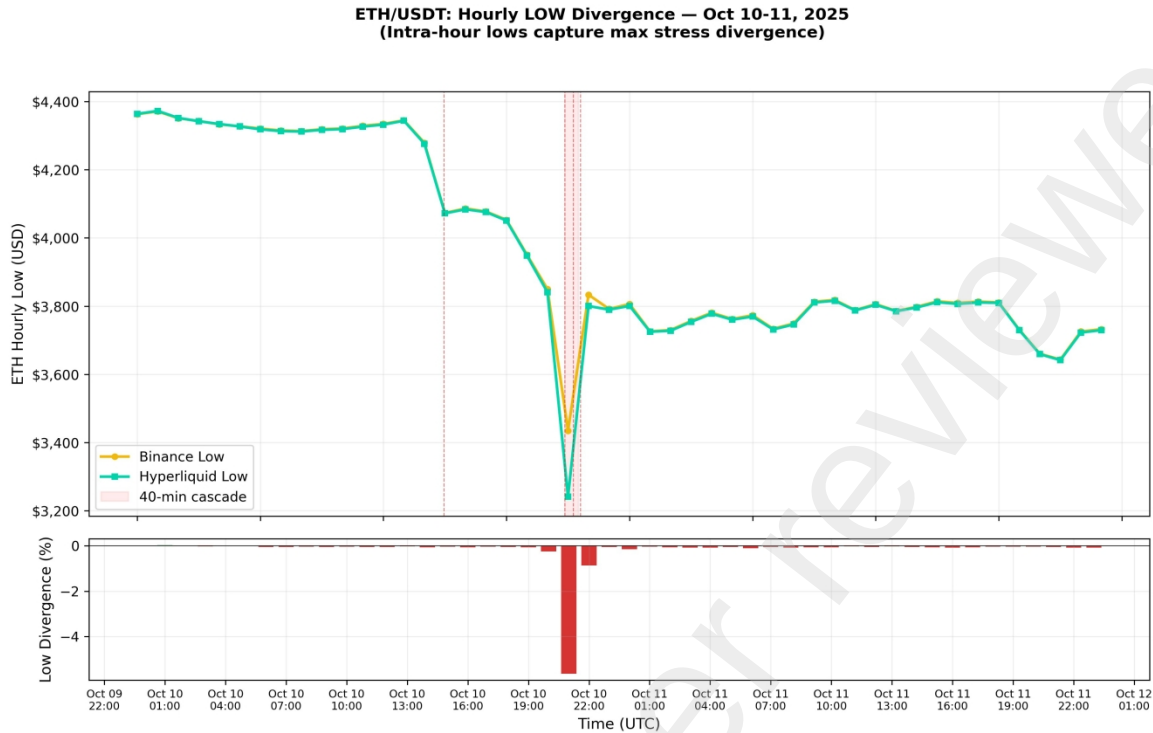
Notes: At tick resolution, BTC divergence reached 7.24% (\$7,387) at 21:24:09 UTC in a single second.

Figure 3. ETH/USDT hourly close: Binance vs. Hyperliquid, October 10–11, 2025



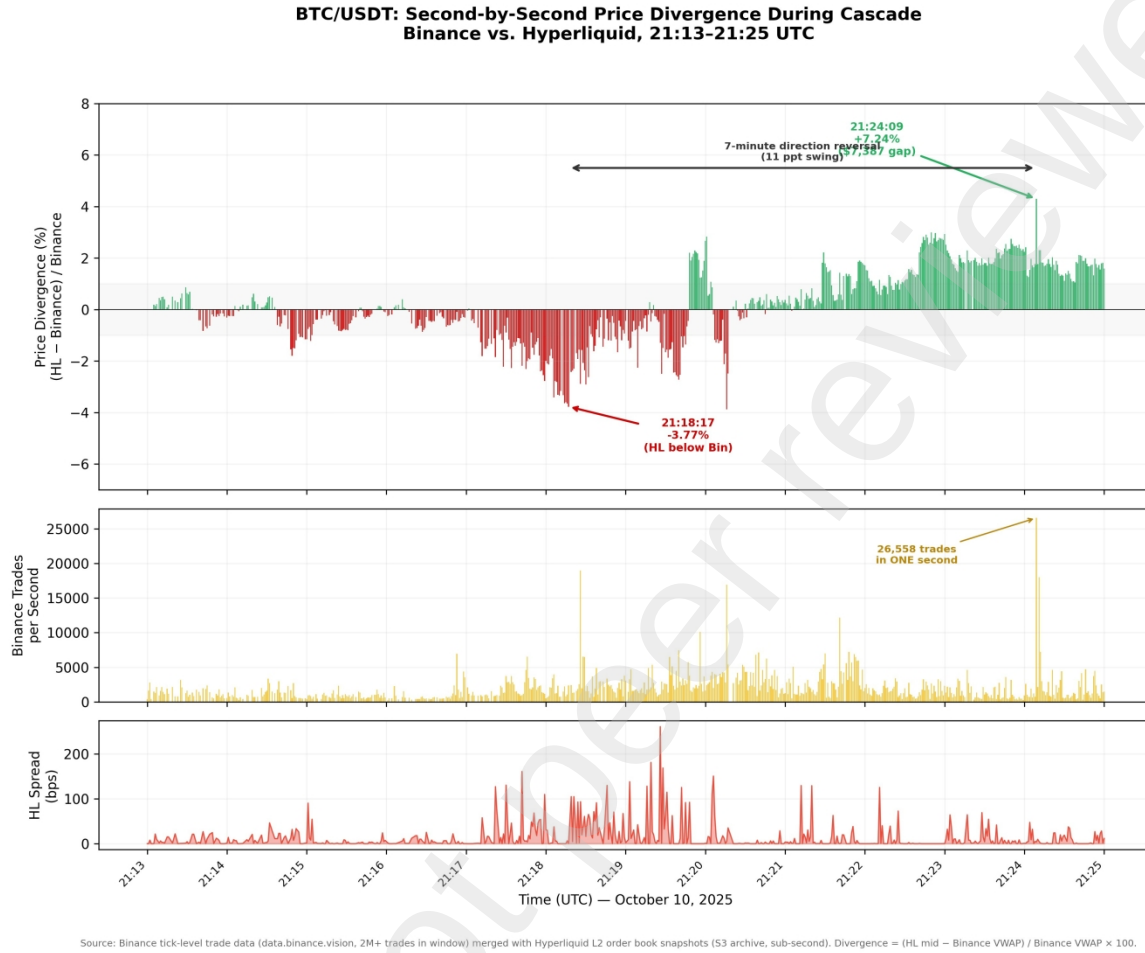
Notes: ETH close-price divergence peaked at 4.76% at 21:20 UTC. Hyperliquid consistently printed lower than Binance on ETH.

Figure 4. ETH/USDT hourly lows: maximum intra-hour stress divergence



Notes: ETH low divergence reached 5.32% (\$185) at 21:19 UTC. Bid-ask spreads widened 602x and bid depth collapsed 99.8% to \$8,745.

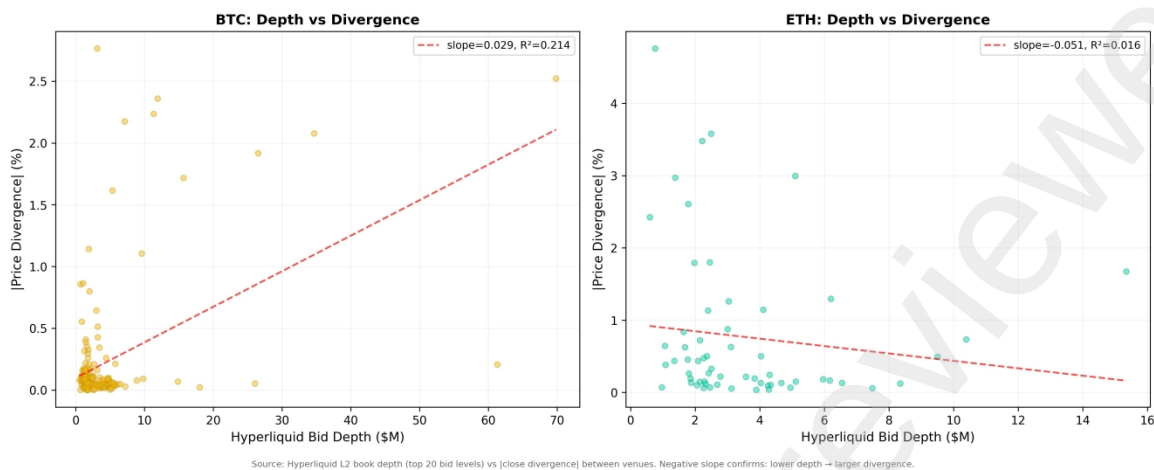
Figure 5. BTC/USDT second-by-second price divergence during cascade (21:13–21:25 UTC)



Notes: Upper panel: per-second divergence. Green = HL above Binance; red = HL below. Direction reversed between 21:18:17 (−3.77%) and 21:24:09 (+7.24%). Middle: Binance trades/sec (peak 26,558). Lower: HL spread (bps).

Figure 6. Order book depth vs. price divergence (mechanism test)

Order Book Depth vs Price Divergence
(Each point = 1 minute during cascade window)



Notes: Each point = 1 minute. ETH shows expected negative slope (lower depth $\rightarrow$ larger divergence). BTC positive slope reflects endogeneity during recovery.

Table 6. OLS regression: |Price divergence| on depth and spread

Variable	Model 1	Model 2	Model 3
Constant	0.2574 (4.99)	0.0571 (1.53)	0.2150 (3.10)
Depth (\$M)	0.0245 (3.78)	0.0228 (5.14)	0.0234 (5.34)
Spread (bps)	—	0.0596 (16.49)	0.0561 (14.82)
BTC dummy	—	—	−0.1980 (−2.69)
R-squared	0.057	0.561	0.574
N	240	240	240

Dependent variable: |Divergence| (%). *t*-statistics in parentheses. Model 2: spread explains 56% of divergence. BTC dummy negative and significant, confirming ETH more fragmented.

Several caveats apply to this regression. First, depth and spread are endogenous—they are outcomes of the same cascade that produces divergence, so the coefficients should be interpreted as associations rather than causal estimates. Second, the model omits DeFi-specific access costs: Ethereum gas prices spiked to approximately 450 Gwei (\$400–500 per transaction) during the cascade, creating a wealth-based access barrier. While DeFi venues maintained 100% uptime, 100% uptime is not the same as 100% accessible uptime. Third, the sample is a single event (N=240 minutes from one cascade), limiting generalizability.

3.7 Robustness checks and extensions

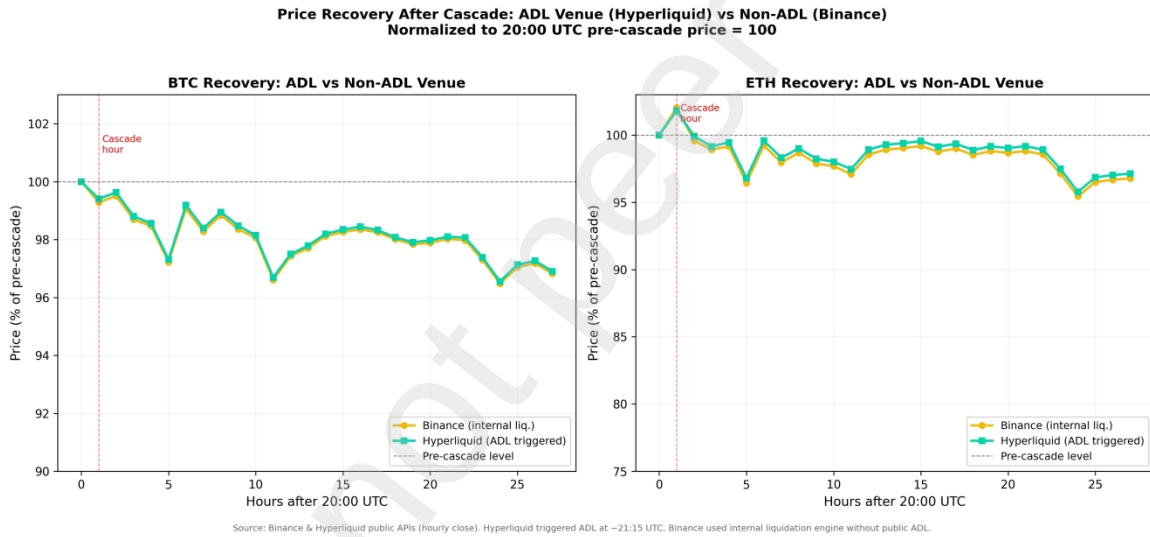
Table 7. Sensitivity of venue liquidation share to underreporting multiplier

Multiplier	True Bin Liq (\$B)	Bin Share	Bin vs HL	Implication
1x (reported)	\$2.4	18.6%	0.2x	HL dominant
5x	\$12.1	53.3%	1.2x	Roughly equal
10x	\$24.1	69.5%	2.3x	Binance dominant
50x	\$120.5	91.9%	11.7x	HL is small fraction
100x (Yan max)	\$241.0	95.8%	23.4x	DeFi negligible

Binance reported \$2.41B; Hyperliquid on-chain \$10.31B; Aave \$0.25B+. The divergence findings (7.24%, 5.32%) are robust to this uncertainty because they use prices, not volumes.

A second concern is whether auto-deleveraging (ADL)—triggered by Hyperliquid at approximately 21:15 UTC—slowed price recovery relative to Binance’s internal liquidation engine. Figure 7 compares hourly close prices on both venues, normalized to the 20:00 UTC pre-cascade price. The result is a null finding: both venues reached their nadir at approximately the same depth (BTC: Binance 96.5%, Hyperliquid 96.6% of pre-cascade; ETH: Binance 95.4%, Hyperliquid 95.8% of pre-cascade) and both recovered to 95% of pre-cascade within two hours. ADL did not measurably impair Hyperliquid’s recovery speed at hourly resolution.

Figure 7. Price recovery after cascade: ADL venue (Hyperliquid) vs non-ADL (Binance)



Notes: Normalized to 20:00 UTC pre-cascade close = 100. Both venues recovered to 95% within 2 hours. ADL did not measurably impair recovery.

Table 8. Extended OLS regression with cascade-window indicator

Variable	Model 4	Model 5	Model 6
Constant	0.0209 (0.54)	0.1559 (2.16)	0.1829 (2.17)
Depth (\$M)	0.0193 (4.28)	0.0203 (4.51)	0.0202 (4.49)
Spread (bps)	0.0548 (14.10)	0.0526 (13.21)	0.0400 (1.96)
Cascade dummy	0.2227 (3.03)	0.1931 (2.60)	0.1712 (2.09)
Spread × Cascade	—	—	0.0129 (0.63)
BTC dummy	—	−0.1632 (−2.20)	−0.1762 (−2.29)

R-squared	0.577	0.586	0.586
N	240	240	240

Cascade dummy = 1 for 20:50–21:30 UTC window. Cascade positive and significant ($t=2.09$ – 3.03). Spread×Cascade insignificant ($t=0.63$): mechanism stable, only intensity changes.

3.8 Price discovery leadership

Following Hasbrouck (1995), we estimate a Vector Error Correction Model (VECM) on second-by-second log prices for BTC/USDT on Binance (tick-level VWAP) and Hyperliquid (L2 mid-price) during the 21:10–21:30 UTC cascade window ($N = 1,102$ matched seconds). We compute both Gonzalo-Granger component shares and Hasbrouck information shares to measure each venue's contribution to permanent price innovation.

A preliminary finding is that the Engle-Granger cointegration test fails to reject the null of no cointegration ($p = 0.61$) during the cascade window, despite both series being individually $I(1)$ with $I(0)$ first differences. This failure of cointegration is itself a finding: under extreme stress, the arbitrage relationship between venues broke down—consistent with our earlier finding that cross-venue arbitrage was non-executable due to transfer delays, subsystem degradation, and depth collapse. The venues were genuinely disconnected.

Table 9. Price discovery leadership: Hasbrouck information shares (BTC, 21:10–21:30 UTC)

Measure	Binance	Hyperliquid
Error correction (α)	−0.0054	0.0175
Gonzalo-Granger share	76.3%	23.7%
Hasbrouck IS (lower)	80.2%	15.2%
Hasbrouck IS (upper)	84.8%	19.8%
Hasbrouck IS (midpoint)	82.5%	17.5%

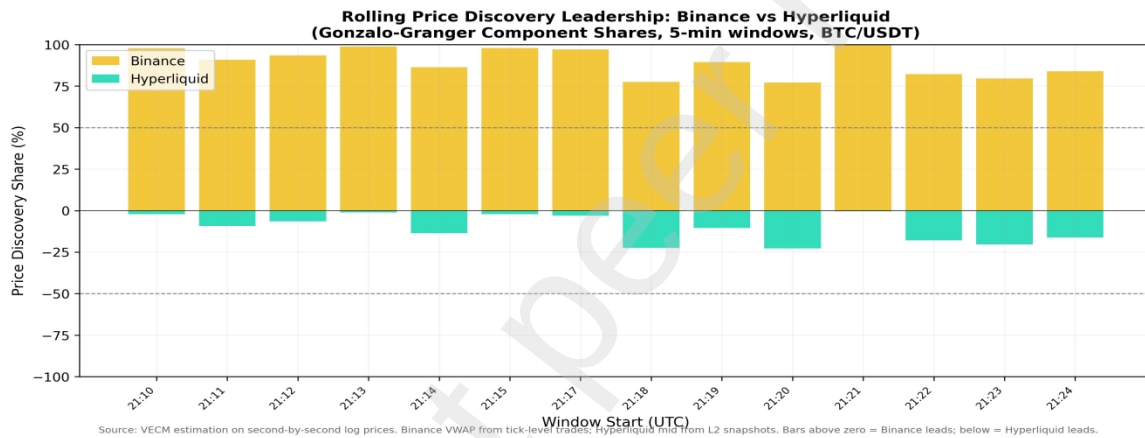
VECM with 4 lags (AIC). 1,102 matched seconds. Binance from tick-level VWAP; Hyperliquid from L2 mid-price. Residual correlation: 0.066.

Binance's error correction coefficient ($\alpha = -0.0054$) is substantially smaller in magnitude than Hyperliquid's ($\alpha = 0.0175$), indicating that Hyperliquid adjusts toward Binance's price 3.2 times faster than the reverse. The Gonzalo-Granger component share attributes 76.3% of price discovery to Binance. The Hasbrouck information share places Binance's contribution at 82.5% (midpoint

of [80.2%, 84.8%] bounds). Binance dominated price discovery throughout the cascade, despite experiencing subsystem degradation.

Figure 8 shows rolling Gonzalo-Granger shares computed over 5-minute windows advanced in 1-minute steps. Binance led price discovery in every single window, with shares ranging from 77.2% to 99.6%. The divergence documented in Sections 3.6–3.7 therefore reflects Hyperliquid’s lagged response to Binance-led price movements, not independent price discovery on Hyperliquid.

Figure 8. Rolling price discovery leadership: Gonzalo-Granger shares (5-min windows, BTC)



Notes: Each bar = 5-min VECM window, 1-min steps. Binance dominated every window (min 77.2%, max 99.6%).

4. Discussion

4.1 The transparency–protection tradeoff

The event crystallized a novel tradeoff in market design. Binance initially reported approximately \$283 million in compensation to affected users, later stating total compensation exceeded \$328 million by October 22, 2025 (Binance, 2026). Hyperliquid compensated nothing—its HLP vault earned approximately \$40 million in liquidation fees that flowed to liquidity providers. Traditional market microstructure theory assumes venues are transparent and available. The October 10 event shows that in crypto markets, transparency and availability are endogenous to venue architecture.

4.2 Regulatory implications

Former CFTC regulator Salman Banaei compared the event to the May 2010 equity Flash Crash, calling for formal investigation. Our analysis suggests three regulatory priorities: (1) standardized liquidation reporting, as the current regime where venues self-report creates systemic opacity; (2) oracle resilience standards, as single-venue oracle dependencies create venue-specific amplification risks; and (3) leverage caps, as the dominance of perpetual futures in crypto trading volume represents a misalignment between platform incentives and systemic stability.

4.3 Limitations and future research

The CeFi data gap—with external analysts arguing underreporting by 10–100x due to stream design limitations—means our analysis necessarily relies on self-reported CeFi figures as a lower bound. Several empirical questions remain open: (1) minute-by-minute price discovery leadership across venues for ETH; (2) effective depth comparison (dollar cost to move price 1%) across CLOBs and AMMs at matched timestamps; (3) whether venues that triggered ADL experienced

slower sub-minute price recovery that our hourly data misses. All three are answerable with existing on-chain data.

It is worth noting that market infrastructure has not remained static since October 10. Binance's January 2026 post-mortem reported that the exchange has since tightened index deviation thresholds, improved cross-exchange reference pricing for collateral assets, and enhanced circuit breakers to reduce recurrence risk (Binance, 2026). Whether these measures are sufficient to prevent a similar cascade remains an empirical question that future stress events will test.

5. Conclusion

The October 10, 2025 cryptocurrency cascade provides a unique natural experiment for comparative market microstructure analysis. We document that identical assets experiencing the same exogenous shock produced radically different outcomes across architecturally distinct venues. The divergence traces to five design choices: oracle sourcing, liquidation transparency, liquidity provision architecture, auto-deleveraging mechanics, and system availability under stress. Empirically, we find that these architectural differences produced extreme price divergence under stress. Using tick-level Binance trade data and sub-second Hyperliquid order book snapshots, we document that at 21:24:09 UTC, BTC prices diverged by 7.24% (\$7,387) between Binance and Hyperliquid in a single second—with 26,558 trades executing on Binance at that moment. The direction of divergence reversed completely within seven minutes as Binance’s internal liquidation cascade pushed BTC below Hyperliquid’s price. ETH diverged by 5.32% (\$185), with bid-ask spreads widening 602x and bid depth collapsing 99.8% to under \$9,000. A Hasbrouck (1995) information share analysis shows that Binance led price discovery with an 82.5% share throughout the cascade—the divergence reflects Hyperliquid’s lagged adjustment, not independent discovery. Our central finding is that market microstructure observability is an architectural property, not a data collection problem. DeFi venues provide complete, verifiable, permissionlessly auditable transaction records. CeFi venues employ liquidation reporting architectures that external analysts have argued understate true volumes by up to 100x during peak stress. This asymmetry means that the true scale of systemic risk during the cascade remains partially unknown—a condition that itself constitutes a form of market risk that current frameworks do not adequately capture.

The event revealed a transparency–protection tradeoff with no clear resolution: CeFi venues compensated losers but obscured the true state of the system; DeFi venues exposed every transaction but offered no recourse. As crypto markets continue integrating with traditional finance, resolving this tradeoff—through circuit breakers, standardized reporting, oracle resilience standards, and informed regulatory frameworks—becomes an urgent priority for market designers and regulators alike.

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